A Decision-making Model for Engineering Designers

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SYNOPSIS

This paper describes research that combines a generic decision-making model, together with design strategies employed by experienced engineering designers. The relationship between the decision-making activities and the eight design strategies are examined. By combining these two understandings of the design process, a model has been developed, which describes the different types of decision-making activities made during the design process together with strategies that aid the designer in making decisions.

1 INTRODUCTION

Engineering design is frequently described as an ill-defined problem: usually many possible solutions exist and there are no clearly defined rules to obtain these solutions [Goldschmidt, 1997; Cross,1997]. Although, there are no clear rules to obtain design solutions, many theories, methods and models of both the *design process* and *technical systems* have been proposed in engineering design. In addition to understanding and supporting the overall design process. Empirical research studies have been carried out to understand the individual designers reasoning process. This paper attempts to bring together these two approaches to understand the process of decision-making and to develop a model to support decision-making in design. The aim of this research was to examine the applicability of a design decision-making model based upon literature to protocols of experienced designers working in the aerospace industry.

1.1 Supporting the design process

Prescriptive methods in engineering design have largely been developed based on the experiences of their authors, and not on observed activities of designers and some validated theory. Until recently, little work has been carried out to evaluate the validity of these methods. Hales, Gunther and Fricke have carried out studies to investigate the validation and usage of some of these design methods [Hales, 1987, Gunther, 1999, Fricke, 1999]. Hales carried out an observation of an engineering design project involving the design of a high-pressure and high-temperature system for testing materials for over two and a half years [Hales, 1987]. He categorised the activities undertaken by the designers and compared these to those suggested by Pahl and Beitz [Pahl, 1996]. Hales found that Pahl and Beitz's categorisation of design phases covered only 47% of the observed activities. The methods and aids described by Pahl and Beitz accounted for 22% of those observed by Hales. A further thirteen categories were identified which accounted for 74% of methods and aids. These categories were grouped under the headings working, communicating and motivating. The additional categories observed by Hales suggest that designers carry out many activities that are not seen as core design activities. Whether, these activities should be included in prescriptive design models is debatable. Currently most models do not include these activities, with the notable exception of Hubka, who identifies them as the 'management' and 'auxiliary' working processes [Hubka, 1982].

Gunther and Ehrlenspiel compared protocols of designers with and without training in design methods [Gunther, 1999]. They found designers with no training: (1) clarified the task throughout the design process rather than the beginning; (2) had shorter design processes overall; (3) produced documentation that focused upon the final design rather than the process; and (4) tended to develop one concept at a time rather than generating many variants. They do not make any conclusions on the *quality* of the design solutions produced. Fricke found that using design methods flexibly, i.e. adapting them to the specific task in hand, has the greatest chance of leading to a successful solution [Fricke, 1999].

In general, prescriptive design methods have given little consideration to redesign work. They have also focused on organising the overall activities of the design process and have not considered the human element or the experience of the designers involved.

1.2 Understanding how individual designers design

Von der Weth states that a designer's knowledge and experience play a significant role when solving problems [Von der Weth, 1999]. In order to support the design process more fully, an understanding of this role and how designers are carrying out design tasks is necessary. Studies of expertise in disciplines such as psychology traditionally deal with well-defined problem areas, e.g. chess playing. The relationship between well-defined problem areas and the ill-defined area of design is poorly understood, and hence the applicability of findings from studies in well-defined areas to design is not clear [Ericsson 1997; Zeitz 1997; Cross, 1997].

Eisentraut found problem solving to be dependant on two things: the characteristic of the individual designers; and the characteristics of the problem [Eisentraut, 1999]. Eisentraut categorises the characteristics of the individual designer in terms of the amount of time spent in the various stages of the problem solving process, including gathering information, carrying out prognosis and planning. He found individuals had developed particular characteristics to solve problems and applied these to all problems. The characteristics of the problem determined the success of these designers.

Fricke's study of nine designers also focused on the time spent at various stages of the design process [Fricke, 1999]. Fricke found that the successful designers clarified and actively searched for requirements. The successful designers did not suppress their first ideas, produced variants and assessed them consciously. Fricke also found successful designers to possess advanced engineering knowledge.
Galle provided designers with a problem brief and a final design solution [Galle, 1996]. The designers were asked to describe the thinking process that they thought would have been used to achieve the final design solution. Galle identified two types of decisions from the analysis: a *that* decision and a *how* decision. A *that* decision specified an end or a goal and a *how* decision identified the means to achieve the end or goal.
Visser and French found that designers either tended to move towards the critical part of the problem or become opportunistic [Visser, 1991, French, 1992]. Stauffer *et al* compared the findings of six empirical studies by separate researchers [Stauffer, 1988]. They concluded that, the actual design process is not sufficiently well organised to be algorithmic as none of the designers observed followed a set procedure. This suggests that designers tend towards a heuristic approach to design.
The literature reviewed suggests that a complete cognitive theory of problem solving in design does not exist and research in this area is still in its early stages. The focus of the research described has been on the external process, such as the time spent gathering information, rather than focusing on the internal problem solving process within a designer's mind, e.g. the time spent on activities, e.g. task clarification and generating concepts [Fricke, 1999]. In addition, the majority of the studies have been conducted in a laboratory environment rather than in industry and, hence, the findings are not necessarily applicable to designers working in industry.

1.3 Decision model

A survey carried out in British industry by Wright et al. [Wright, 1995] indicates that design methods are sparsely adopted and used in industrial practice. With respect to decision-making during the design process this is an unfortunate situation for two reasons. Firstly, because of the importance of making the right decisions during design, and secondly because many methods to support decision-making do exist, see e.g. [Jones, 1970], [Pahl, 1996], and [Roozenburg, 1995].
From the literature study it is clear that there are many methodologies and guidelines to support decision-making in design, and interesting elements of insight from empirical studies of decision-making are reported. However, the area has few fundamental works, which could serve as a theory base. Further, it is observed that the lack of a functional and comprehensive mindset for decision-making in engineering design seems to have resulted in a situation, where existing methods are only sparsely used in practice. The literature review led to the development of a decision-making model *the decision node* of Hansen, which was based on decision-making and engineering design theories [Hansen, 2000]. The *decision node* is a model of the interrelated decision-making activities when making a decision during design. The *decision node* is a generic, elementary decision-making activity consisting of six sub-activities: to *specify*, to *evaluate* solution alternatives, to *validate* a design solution, to *navigate* through the solution/activity space, to *unify* the current decision into consistent wholes, and to *decide*.

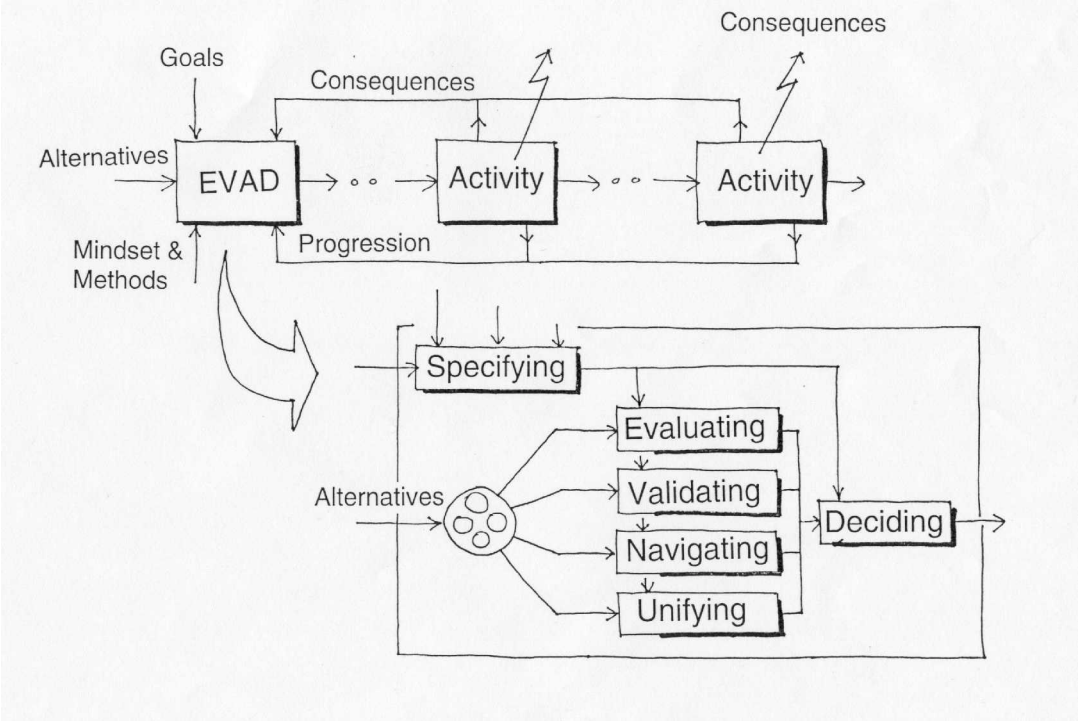


Figure 1 Decision-making model

1.4 Strategies employed by experienced designers

Eight design strategies employed by experienced designers were identified from empirical research carried out within the aerospace industry. Twelve observations together with thinking-aloud were used to understand the differences between how novices and experienced designers approached their design tasks [Ahmed, 2001]. The transcripts were analysed and a total of twenty-two categories describing the thoughts and actions of the designers were generated. Eight of these categories have been identified as design strategies, these were observed to be predominantly experienced designer behaviour. The strategies were:

- **Consider issues:** the experienced designers tended to consider several relevant issues, and decided which were the most important. They were also aware when issues were not relevant. Example questions: Is there a problem with welding or machining that?
- **Question data:** the experienced designers questioned data they obtained from any source. This included values given for tolerances, standards, stress models, etc. They questioned the accuracy of the data; how components were modelled or tested; how much accuracy was required; customer specifications; and the applicability of standards. Example included: How does the test data from this engine relate to how it would behave in the real world?
- **Question is it worth pursuing:** the experienced designers asked themselves how much they could expect to achieve if they continued a particular approach (i.e. if they implemented a decision) and if it was worthwhile. Example questions: How many engines with nose cones are there and is it worth pursuing for that number of engines? Can we achieve those cost savings?
- **Aware of reason:** the experienced designers were often aware of the reasons behind the use of a particular component (or assembly or feature) or manufacturing process in a particular design. The reasons why a component or process was used may be due to a specific function or the capability of a particular supplier or manufacturing process. In the

latter case, knowledge of previous projects may be required (i.e. refer to past designs). The experienced designers assessed the reasons and their applicability in the current situation. They often referred back to past projects to compare how a component behaved previously. If removing or modifying a component or process would help solve the design problem, this would be carried out. Example questions**:** Why do engines use that titanium?

- **Aware of limitations:** the experienced designers were aware of the limitations of the current design task and hence of the amount of time to spend on it. The following reasons were identified to limit the task: the expected achievement of the current task versus further design tasks; and incompleteness of information e.g. if further changes were required or components needed testing. Example questions: What are the limitations of using heat seals, their testing doesn't include heat?
- **Aware of trade-offs:** the experienced designers were aware of the relationships between issues. When considering an issue they knew the effect this issue had on other issues. They were aware that many decisions were based on compromises. Once aware of the trade-off, they would question whether it was worthwhile continuing to pursue the task or implementing a decision. Example questions: If the performance is good enough is the extra cost ok?
- **Refer to past design:** the experienced designers referred to past projects to find similar designs; designs in similar environmental and functional conditions; and where similar problems had been encountered and how they were resolved. Example questions: Has anything like this been done before?
- **Keep options open:** the experienced designers rejected an option or delayed a decision on an option if it limited later options in the design task. They were aware of what needs had to be considered further down the design process. Example questions: Are there any other problems that we can address whilst making a change?

2 HYPOTHESIS AND RESEARCH APPROACH

Two of the protocols used to identify the design strategies described above were reanalysed to understand the applicability of the design decision-making model. Both participants were experienced designers working in the aerospace industry on real design tasks. The observations were around ninety minutes in duration and were followed by a short interview to understand the background of the designer and design task, these have been summarised in Table 1.

Designer	Years of experience	Stage of Design process	Project time in weeks		Task description
			Time already spent on project	Total expected time of project	
A	19	Detailed Design	56	64	Design of a shaft
B	8	Detailed Design	52	54	Design O-ring and squeeze film

Table 1 Background of design task and designer

The main objectives of this analysis were:

- To understand if these decision-making activities existed during the protocols?
- To identify additional activities that occurred during the protocols to those described by the model?

- To understand if the strategies (described in section 1.2) were related to the decision-making activities.

2.1 Hypothesis

It was hypothesised that:

- the analysis of the protocols would identify the decision-making activities proposed in the model, but not necessarily all of them.
- as the decision-making model proposed is based on literature of prescriptive methods, and these methods are generally not based upon empirical research of individual designers, it was expected that the analysis of the protocols would identify activities that could not be explained by the decision-making model.

2.2 Research method

The analysis of the two protocols was carried out in two phases:

Phase 1: During the *encoding phase* the decision-making episodes were identified. Many decisions occur during the design process, the level at which a decision is identified was important for this research. A decision could be defined at many different levels, for example:

1) the decisions involved in the selection of an overall concept;
2) e.g. the decision to select the features of the concept;
3) and it could be argued that the process of drawing one concept involves many decisions for each line of the process of sketching [for example the research of Suwa, 1997].

For the purpose of this paper the decisions identified were defined as a requiring a conscious thought process. As protocols with thinking-aloud were used, the decisions identified were limited to those the designer verbalised. Each of decision-making episodes was encoded into an encoding scheme, the encoding scheme used to identify the decision-making sub-activities were:

- *Specify*: a statement concerning compilation of design criteria
- *Evaluate*: a statement concerning either the value of a design alternative, on a design alternative being better/worse.
- *Validate*: a statement whether a design alternative is "fit for purpose", e.g. "not problematic", *purpose* includes consideration of the product lifecycle, e.g. consider the use, manufacture, etc.
- *Navigate*: a statement regarding the progression and feasibility of the design work, i.e. which activity to do next or in which direction to go next
- *Unify*: a statement concerning the current design solution or design activity in relation to the totality of the product or process
- *Decide*: a verbally expressed decision
- *Other*: statements which do not belong to any of the first six categories

Phase 2: During the *analysis phase* the encoding schemes were analysed. The encoding schemes were then compared with a previous analysis to identify the occurrence of the eight design strategies. The previous analysis used an encoding scheme consisting of twenty-two categories describing the designer's thoughts and actions, however the research was focused on a subset of eight of these categories that described design strategies (described in section 1.2) and not the remaining categories. The encoded scheme used was: consider issues, question data, question is it worth pursuing, aware of reason, aware of limitations, aware of trade-offs, refer to past designs and keep options open. The protocols had already been encoded, and therefore, care was taken to avoid influencing the analysis by not referring to this encoding until phase 1 was completed. During this phase the protocols were analysed to

identify any relationships between the design strategies and the decision-making sub-activities. Observations of any patterns in the order of which the strategies were used during the decision-making sub-activities were formulated.

3 FINDINGS

3.1 Decision-making activities

The protocols were divided into sequences, the end of the sequences corresponded to a decision being made. The two authors independently identified the sequences and were in full agreement. Five sequences were observed during the protocol of designer A and, four with designer B. During the two protocols, all of the five decision-making activities were observed (refer to Table 2). However, there was a difference in these activities, only evaluate and validate were observed as decision-making activities, i.e. the entire purpose of the sequence was to make a decision to evaluate between alternatives or to validate the design. The remaining three activities: to specify, to unify and to navigate were observed as part of a validating or evaluating sequence. These three activities were never observed on their own, and hence suggest that the designers used these three activities to guide them in making a decision *about* validating the design or evaluating the design. In addition, to these activities both designers spent periods commenting on the design problem, which may have been for the benefit of the researcher, these periods have been excluded from the analysis. Designers A was interrupted by two telephone calls during the protocols, these periods have also been excluded from the analysis.

Activity	Overall sequence	Designer A	Designer B
To specify	validate, evaluate	2	1
To evaluate	evaluate	1	1
To validate	validate	2	3
To unify	other	1	0
To navigate	validate	0	2
other	other	2	0

Table 2 Decision-making activities observed

3.2 Additional activities observed

In total, nine sequences were observed, two of these were encoded as *to evaluate* and five as *to validate.* The remaining two sequence were described as *other,* one of these was the start of a sequence of activities, however the period of observation ended too soon to be able to encode the sequence. The final sequence categorised as *other* was during the protocol of designer A. The designer was working on the design of shaft and had been informed by the manufacturers that there was a problem in manufacturing the locking ring on the shaft. The manufacturers requested that the tolerances be relaxed as this would save time and cost. Hence, the validation of the design had already taken place, the manufacturers had decided that the design was not fit for purpose, in that it was difficult to manufacture. The designer had to decide whether the tolerances on the locking ring could be relaxed, this is neither a validating or evaluating activity (or any of the other decision-making activities). During this sequence the designer is describing his thought process which led him to decide that the tolerances could be relaxed. This sequence is a decision-making activity that is not described

by the model and is a result of a validation being negative. During this sequence, the designer was observed to use design strategies as described in section 1.2.

3.3 Relationships between strategies and decision-making activities

During the protocols, the designer was observed to be using the design strategies whilst evaluating or validating. Each of the evaluating and validating sequences are described together with the decision-making activities and design strategies employed in Table 3. All eight of the design strategies were observed during these sequences.

Designer	Overall decision-making activity	Description of sequence	Decision-making activities observed during sequence	Design Strategies
A	validating	Is this design drawing fit for purpose?		refer to past designs
			Specifying	consider issues
				refer to past designs
A	validating	Is specification of shaft suitable for manufacturers?		refer to past designs
A	evaluating	Should the design have a cold-expansion process on the holes? Evaluating between a design with process and without		consider issues-aware of reason-refer to past designs
				aware of limitations-aware of reason-question is it worth pursuing -refer to past designs
				consider issues-refer to past designs
B	validating	Is this O-ring fit for purpose?		consider issues-question data
				consider issues
			Navigate	question is it worth pursuing
B	validating	Does the length of squeeze film need to be extended or is it good enough?		aware of reason
				aware of limitations-question data-aware of reason
				consider issues
			Navigate	consider issues-question is it worth pursuing-keep options open
				question data
B	Evaluating	What are the alternative ways to improve the film?		consider issues
				consider issues-aware of reason
			Unify	aware of trade-offs
				aware of reason
				question data
			Specify	consider issues-refer to past designs
				consider issues-aware of reason -question is it worth pursuing

Table 3 Decision-making activities and design strategies

All of the design strategies were observed during the decision-making activities *to evaluate* and *to validate.* The strategy *question is it worth pursuing* was used by the designer before

making a decision in all but two sequences. Both these sequences were validating the design drawing and hence, the decision to pursue the design has already been made. Therefore, *question is it worth pursuing* seems to be used by the designers just before validating or evaluating a design. The activity *to specify* seems to be related to the design strategy *consider issues. To specify* only appears with consider issues, however *consider issues* appears many times without the activity *to specify* and hence, suggests that *consider issues* (CI) is a strategy that aids the designer *to specify* as well as other activities. The activity *to navigate* always appears together with *question is it worth pursuing* (WP). This strategy is related to the design process, as is *keep options open* (OO) and hence, it was expected that these strategies are related to the activity *to navigate.* The activity *to unify* was observed only once, in this instance, the activity was used together with *aware of trade-offs* (TO) to consider the effects of the choice of alternative.

The findings are based upon the analysis of two protocols and within a particular company within the aerospace industry hence and therefore conclusions that can be drawn are limited. Although, analysis of further protocols are planned.

4 DECISION-MAKING MODEL

The process of analysing the data found that the decision-making model can be separated into two levels. The first level is the decision-making during the overall design process; and the second level is concerned with the reasoning of individual designers when making decisions. The decision-making model at the individual designer level has two decision-making activities: to evaluate and to validate (refer to Figure 2). The design strategies together with *to specify*, *to unify* and *to navigate* are signals that aid the designer during these activities.

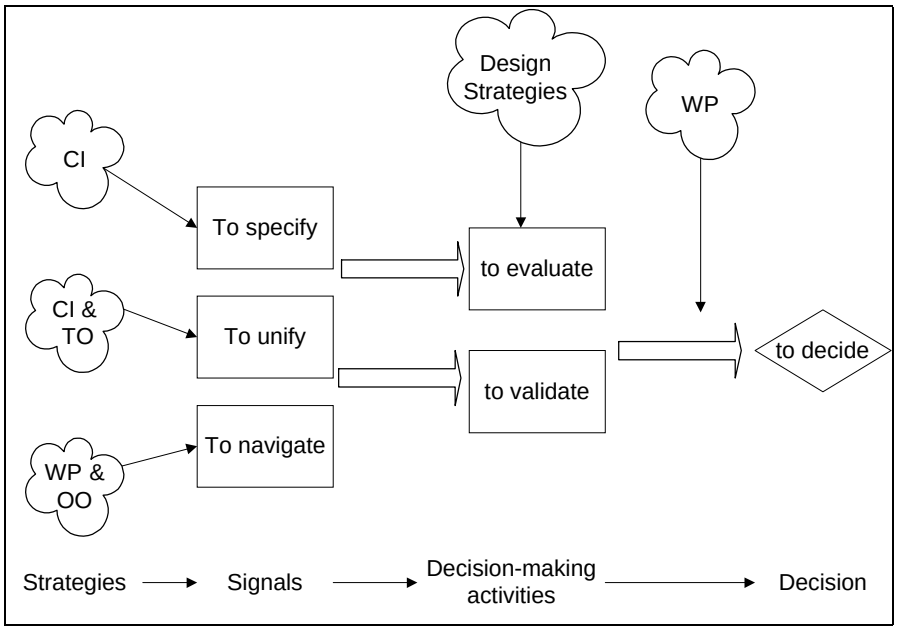


Figure 2 Decision-making model: Individual designer level

CI: consider issues, TO: aware of trade-offs, WP: question is it worth pursuing, OO: keep options open

The decision-making model for the overall design process remains the same as Figure 1, although the activities *to unify* and *to navigate* were not observed as separate decision-making

activities during the protocols, these activities may still exist during the overall design process. The protocols represent only a short episode of the design process, and these decision-making activities may have been observed if protocols had taken place at a different time.
In addition, to these differences, there were also differences in *how* the designers were evaluating designs and what is generally understood as evaluating between concepts. Pugh states that when evaluating between alternatives, all the alternatives should be developed to the same level and implies that the alternatives are represented in an external form [Pugh, 1997]. However, during the protocols both designers were observed to evaluate designs that had yet to be externalised. The ability to evaluate designs before they are externalised is attributed to experienced rather than novice designers [Ahmed, 2001]. The designers did not externalise their design alternative unless their evaluation had been successful, if not they would generate another design alternative. Hence, the evaluation was between alternatives, but was sequential. One limitation of thinking-aloud protocols is that the designers can only verbalise their ideas sequentially.

4 CONCLUSIONS

A decision-making model has been developed through applying a generic model with protocols of experienced designers working in industry. The analysis of the protocols has led to a deeper understanding of the decision-making activities undertaken by designers within an industrial context. The decision-making activities undertaken by individual designers were supported by design strategies. This implies that designers in industry do not rely solely on decision methods to support their decision-making process, but also on the use of relevant design strategies. This finding has implications to teaching design decision-making methods to engineering students and designers working in industrial practice: in addition to teaching decision-making methods and techniques found in literature, we must also include a description of decision-making activities together with a set of relevant strategies based upon empirical research.

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6 REFERENCES

[1] Ahmed S (2001) *Understanding the Use and Reuse of Experience in Engineering Design*. Cambridge University, Ph.D. thesis, Cambridge

[2] Cross N and Cross A C (1997) Expert Designers. In: (eds) *Designers-The key to Successful Product Development.* Springer-Verlag, Darmstadt, pp 1-16

[3] Ehrlenspiel K and Dylla N (1989) Experimental Investigation of the Design Process. In: (eds) *International Conference on Engineering Design.* St Edmundsbury Press, Harrogate, pp 77 - 95

[4] Eisentraut R (1999) Styles of Problem Solving and their Influence on the Design Process. *Design Studies* 20:431-437

[5] Ericsson K A and Charness N (1997) *Cognitive and Development Factors in Expert Performance*. In: J.Feltovich P, K M Ford and R R Hoffman (eds) Expertise in Context. AAAI Press, California, pp 3-41

[6] French M (1992) The opportunistic route from function to form. In: *EDC Function to Form workshop*. Cambridge, pp 16-20

[7] Fricke G (1999) Successful Approaches in Dealing with Differently Precise Design Problems. *Design Studies* 20:417-429

[8] Galle P and Kovacs L (1996) Replication Protocol Analysis: a Method for the Study of Real-World Design Thinking. *Design Studies* 17:2:181-200

[9] Goldschmidt G (1997) Capturing Indeterminism: Representation in the Design Problem Space. *Design Studies* 18:4:441- 455

[10] Gunther J and Ehrlenspiel K (1999) Comparing Designers from Practice and Designers with Systematic Design Education. *Design Studies* 20:439-451

[11] Hales C (1987) *Analysis of the Engineering Design Process in an Industrial Context.* University of Cambridge, Ph.D thesis, Cambridge

[12] Hansen C T and Andreasen M M (2000) Basic thinking patterns of decision-making in engineering design. International Workshop on Multi-criteria Evaluation, MCE 2000 Neukirchen, pp 1-8

[13] Hubka V (1982) *Principles of engineering design*. Butterworth & Co (publishers) Ltd., Trowbridge, pp. 112

[14] Jones J C (1970) *Design Methods – seeds of human futures*. Wiley-Interscience, John Wiley & Sons Ltd., London

[15] Pahl G and Beitz W (1996) *Engineering Design*. Springer-Verlag, London,

[16] Pugh S (1997) *Total Design: Integrated Methods for Successful Product Engineering.* Addison-Wesley Publishers Ltd., Wokingham, pp 267

[17] Roozenburg N F M and Eekels J (1995) *Product Design: Fundamentals and Methods.* John Wiley & Sons, Chichester, pp 408

[18] Stauffer L A and Ullman D G (1988) A Comparison of the Results of the Empirical Studies into the Mechanical Design Process. *Design Studies* 9:2 April:107 - 114

[19] Suwa M and Tversky B (1997 October) What do Architects and Students Perceive in their Design Sketches? A Protocol Analysis. *Design Science* 18:4:385 - 403

[20] Visser W (1992) Designer's Activities examined at three Levels: Organisation, Strategies and Problem-solving Processes. *Knowledge Based Systems* 5:1:92-104

[21] Weth R V D (1999) Design Instinct? The Development of Individual Strategies. *Design Studies* 20:453-463

[22] Wright I C, Campello A C et al. (1995) A Survey of Methods Utilisation During Product Design Process in UK Industry. Engineering Design Institute, Loughborough University of Technology

[23] Zeitz C M (1997) Some Concrete Advantages of Abstraction: How Experts' Representations Facilitate Reasoning. In: J.Feltovich P, K M Ford and R R Hoffman (eds) *Expertise in Context*. AAAI Press, California, pp 43-65